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What the data platform needs to reach outside the network

Doubtless data platform, Germany West Central

Cynozure

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Purpose

This note sets out what the data platform needs to reach outside the Doubtless network, and why. It is written from the data platform side. It brings together Microsoft's published requirements for Azure Databricks and what our deployment code actually does, so that the network low-level design has a clear starting point.

It is an input to that design, not a firewall specification. The routing, the type of each rule and the final rule set are Hexaware's to decide and own. We would expect them to check what follows against Microsoft's current documentation and their own standards, and to tell us where the design will work differently.

How the platform is set up

A few choices in the platform design keep the outbound list short. The workspaces run classic compute only (serverless compute is parked), injected into the Doubtless VNets, with no public IP addresses on cluster nodes. Public network access to the workspaces is switched off, and each workspace reaches the Databricks control plane through its own private endpoint. Our storage accounts, both Key Vaults and the workspaces' own storage are also reached through private endpoints.

That means most of the platform's traffic stays inside the VNet. What is left is traffic to services Databricks runs for itself, and the traffic our deployment pipelines need.

Each destination below is given two ways: by hostname, and by the Azure service tag that covers it. The port is the same either way. We checked each tag on 16 October 2026 by resolving the hostname and matching its address against Microsoft's published service tag file (version dated 12 October 2026).

1. Databricks compute

The source is the host and container subnets of each workspace: two pairs per environment, one for the analytical workspace and one for the operational workspace. Clusters and the SQL warehouses both run there.

Our code adds no libraries, start-up scripts or package repositories to clusters, so everything here is something Databricks itself needs to start and run a cluster.

#	What it is for	Hostname	Service tag that covers it	Port
C1	Databricks runtime images and libraries (artifact storage)	`dbartifactsprodgerwescen.blob.core.windows.net`	`Storage.GermanyWestCentral`	TCP 443

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#	What it is for	Hostname	Service tag that covers it	Port
C2	Second copy of the artifact storage	`dbartifactsprodswissnor.blob.core.windows.net`	`Storage.SwitzerlandNorth`	TCP 443
C3	Cluster logs	`dblogprodgerwescen.blob.core.windows.net`	`Storage.GermanyWestCentral`	TCP 443
C4	Databricks system tables	`ucstprdgerwescen.dfs.core.windows.net`	`Storage.GermanyWestCentral`	TCP 443
C5	Databricks service logging (Event Hubs)	`prod-germanywestcentral-observabilityEventHubs.servicebus.windows.net`	`EventHub.GermanyWestCentral`	TCP 9093

C1 matters most. Every cluster and SQL warehouse downloads its runtime image from here when it starts, so nothing runs without it. The images are large, and Microsoft warn that if the download is slowed, cluster creation fails with the error `SLOW_IMAGE_DOWNLOAD`.

C2 is easy to miss. Databricks keep the second copy of Germany West Central's artifact storage in Switzerland North, so a rule set built only from Germany West Central tags will not cover it.

C5 is not HTTPS. C5 uses the Kafka protocol, which may affect how those rules can be written.

What we believe is not needed

The `AzureDatabricks` service tag, which covers the Databricks control plane and web application, should not be needed from these subnets. Clusters reach the control plane through the workspace's private endpoint instead. Microsoft say so directly: 'If Azure Private Link is enabled on your workspace, the Azure Databricks service tag isn't required.'

Public package repositories (PyPI, Maven, CRAN) are not needed to build or test the platform. Data engineers are likely to ask for them later, and that is a separate decision for Doubtless security.

Clusters signing in to Entra ID directly (`login.microsoftonline.com`, tag `AzureActiveDirectory`) is only needed if notebook code authenticates to Azure services with its own credentials. Unity Catalog does not need it, so we suggest leaving it out for now.

2. Hostname or service tag

Both will work, and the choice is Hexaware's to make with the Doubtless security team. The point we would raise, from a data protection angle, is what each one lets through.

A service tag is a list of Microsoft IP ranges for a whole Azure service, kept up to date by Microsoft. `Storage.GermanyWestCentral` covers every storage account in the region, not only the Databricks ones, and the same goes for the Event Hubs and SQL tags. Microsoft's own guidance notes that the tags 'cover the entire Azure Storage and Azure Event Hubs services,t only the Azure Databricks endpoints.' So with tags, code running

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on a cluster could copy data to a storage account outside Doubtless's control, as long as it sits in the same region.

Hostnames close that gap, because clusters can then reach only the named Databricks accounts. The cost is upkeep. If Databricks add a hostname for the region, clusters fail until the list is updated. This does happen: some other regions already spread their artifact storage over as many as 15 accounts. Germany West Central has one today.

One thing to add here. IPH in the US use service tags in their firewall for the Databricks platform they host. Calandra operates a different network structure, and we have tighter data sharing rules in the UK and Europe, so both of these things might make the tag route impossible. We wanted to flag this however as a decision point.

3. Build agents

The source is the agent pool our pipelines run on, `dpx_inf_gwc_linux_pba`. Production will have its own pool, which is not named yet, and will need the same list.

Our plan rehearsal runs on 15 October already went through this pool, so most of this list is open today. The one we know is missing is A1.

#	What it is for	Hostname	Service tag that covers it	Port	Status
A1	Databricks account API, used to set up the account-level parts of the platform	`accounts.azuredatabricks.net`	`AzureDatabricks`	TCP 443	Blocked: our run on 13 October was cut off during connection set-up
A2	Databricks Terraform provider download	`registry.terraform.io`, `github.com`, `objects.githubusercontent.com`, `release-assets.githubusercontent.com`	None	TCP 443	Not yet tested
A3	Sign-in for the pipeline identity	`login.microsoftonline.com`	`AzureActiveDirectory`	TCP 443	Working
A4	Azure Resource Manager, used to deploy the Azure resources	`management.azure.com`	`AzureResourceManager`	TCP 443	Working
A5	Terraform itself	`releases.hashicorp.com`	None	TCP 443	Working
A6	Terraform providers and Azure Verified Modules	`registry.terraform.io`, `releases.hashicorp.com`, `github.com`, `codeload.github.com`	None	TCP 443	Working
A7	Tools installed at run time if missing (Azure CLI, unzip, a Python library)	`packages.microsoft.com`, the Ubuntu package mirrors, `pypi.org`, `files.pythonhosted.org`	None suitable	TCP 443, and 80 for some mirrors	Working

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A1 is the one that blocks us now. The Databricks account API has no private endpoint, so the pipeline can only reach it over the internet. The hostname directs each caller to its nearest Databricks region; from Germany West Central it lands on an address inside the `AzureDatabricks` tag.

A2 has not been tested because the Databricks pipeline has not yet run. The GitHub hostnames are where the provider is normally published, not something we have watched it fetch.

A7 would go away entirely if the agent image came with these tools already installed. For `packages.microsoft.com` the only covering tag is Azure Front Door's, which covers every Front Door customer, so we have marked it 'none suitable'.

The agents also need what every Azure DevOps agent needs, and the storage account that holds Terraform state. Both work today.

The pipelines also call each workspace directly, at addresses of the form `adb-<id>.<n>.azuredatabricks.net`. That is not internet traffic. Those names have to resolve to the workspace's private endpoint, and the agents need a route to it. Our pipeline checks this before it starts and stops if a workspace resolves to a public address.

4. Things we have noticed that affect this

These are observations from our side. How to handle them is for the network design.

In the landing zone code as it stands, the compute subnets have outbound access switched off and their route tables hold no routes. Until an outbound route exists, no cluster or SQL warehouse can start. We assume that will come with the hub connection.

Clusters also talk to their own workspace's private endpoint inside the VNet, on TCP 443, 6666, 3306 and 8443 to 8451, and to the storage and Key Vault private endpoints on 443. None of that needs the internet. If traffic within the VNet is routed through the firewall, though, it would need allowing there too.

Microsoft document an alternative for the artifact storage traffic: a storage service endpoint on the compute subnets, with a policy limited to Databricks' own storage. It keeps the large image downloads off the firewall. We mention it for Hexaware to weigh up; we have not assessed it against the landing zone's policies.

The first sign of a gap is likely to be the Databricks deployment itself. It creates a SQL warehouse that starts straight away, and if C1 to C5 are not reachable the deployment will wait and then time out.

5. How we will check it from our side

Once the rules are in place in development, we will run the Databricks pipeline. Its first step checks the workspace address resolves privately and calls the account API, which proves A1. We will then start a cluster and run a query. If the cluster fails, its error tends to point at the cause: 'Control Plane Request Failure' suggests the private endpoint or DNS, `SLOW_IMAGE_DOWNLOAD` suggests the artifact storage path, and codes starting `X_NHC_` name a failed DNS check. We will share whatever we find.

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Sources

- Microsoft, IP addresses and domains for Azure Databricks services and assets (updated 13 October 2026): https://learn.microsoft.com/en-us/azure/databricks/resources/ip-domain-region
- Microsoft, User-defined route settings for Azure Databricks (updated 11 September 2026): https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/udr
- Microsoft, Configure classic compute plane private connectivity (updated 9 October 2026): https://learn.microsoft.com/en-us/azure/databricks/security/network/classic/private-link-standard
- Microsoft, Disable access to the Hive metastore: https://learn.microsoft.com/en-us/azure/databricks/data-governance/unity-catalog/disable-hms
- Microsoft, Azure IP Ranges and Service Tags, Public Cloud (file dated 12 October 2026): https://www.microsoft.com/en-us/download/details.aspx?id=56519

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